

StatLM: A Hybrid Pipeline for Zero-shot Vietnamese Multi-Label Text Classification

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Abstract—The rapid growth of digital documents has increased the need for scalable document organization across languages and domains. Open-world zero-shot multi-label text classification is challenging because neither the label space nor the document-level labels are specified in advance, and task-specific labeled data are unavailable. Existing approaches typically trade off label-space quality, large language model (LLM) usage, and processing time. This paper presents StatLM, a hybrid pipeline that combines a statistical encoder with an LLM decoder. The encoder first applies extractive TextRank using Vietnamese sentence embeddings and then uses KeyBERT with an asymmetric retrieval model to represent each document by a compact set of keyphrases. The LLM is invoked only at the final decoder stage to construct and normalize the label space and to assign labels. On a balanced VietnamNet corpus of 2,000 articles from 20 categories, StatLM increases the coarse-label ratio from 32.6% to 76.9%, reduces the noise-label ratio from 39.1% to 3.8%, and covers 19 of 20 categories, compared with 15 of 20 for MiniLM, our X-MLClass-based full-LLM baseline. StatLM also reduces the mean processing time from 40 to 10 seconds per document, uses approximately 30 times fewer LLM tokens, and requires one API call rather than approximately eight. These results indicate that statistical noise reduction and content condensation before LLM-based label generation provide a favorable quality, efficiency trade-off for resource-constrained deployment.

Index Terms—Coarse labeling, hybrid pipeline, KeyBERT, large language model, statistical encoder, StatLM, TextRank, Vietnamese natural language processing, zero-shot multi-label text classification.

I. INTRODUCTION

The volume of Vietnamese digital text from online news, electronic commerce, social media, and academic repositories continues to grow, making manual document organization increasingly difficult. This study considers *open-world zero-shot multi-label text classification*, in which a system must infer the label space and assign one or more labels to each document without task-specific labeled data or a predefined label inventory [1]. In this setting, the label space may be expanded as additional document batches are processed. Such a capability is relevant to document collections for which a fixed taxonomy is unavailable or expensive to maintain. Existing methods can be broadly grouped into three categories [1]. Supervised classifiers can achieve high predictive accuracy when a complete label set and sufficient labeled examples are available, but these assumptions do not hold in the target

setting. Direct LLM-based methods can generate labels from raw text without task-specific training, but repeated processing of long documents increases token usage and latency and may produce entity-level or overly specific labels. Statistical methods, including topic modeling and keyphrase extraction, are computationally efficient but generally return topics or phrases rather than a normalized, human-readable multi-label taxonomy. These limitations motivate a method that combines statistical filtering with LLM-based semantic generalization.

We propose *StatLM*, a pipeline comprising a statistical encoder and an LLM decoder. The encoder removes low-centrality content and represents each document by a compact, diverse set of keyphrases. The decoder then uses these representations to discover, normalize, and assign coarse-grained labels. Restricting LLM processing to the final stage is intended to retain the semantic flexibility of generative models while reducing exposure to raw-text noise and limiting API usage.

The contributions of this work are as follows. First, we formulate a hybrid encoder-decoder architecture for open-world zero-shot multi-label classification and position it relative to supervised, statistical, and direct LLM-based alternatives. Second, we develop a Vietnamese processing pipeline that uses embedding-based TextRank for extractive summarization and configures KeyBERT with an asymmetric retrieval model for keyphrase extraction. Third, we construct and release a balanced VietnamNet corpus and evaluate the proposed pipeline at the summarization, keyphrase-extraction, label-space, classification, and computational levels against statistical baselines and an X-MLClass-based full-LLM baseline.

The remainder of this paper is organized as follows. Section II reviews the relevant methods and Vietnamese language resources. Section III describes the StatLM architecture, and Section IV presents the dataset, baselines, and evaluation protocol. Section V reports and discusses the experimental results. Section VI summarizes the findings, limitations, and directions for future work.

II. BACKGROUND AND RELATED WORK

A. Graph-Based Extractive Summarization

Extractive summarization preserves source wording and requires no summary-specific training data. Graph-based methods represent sentences as nodes and their relations as

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weighted edges, then estimate sentence importance from the graph structure. TextRank applies PageRank to this representation [2], but its performance depends on the edge-weight function. Word overlap and variants based on TF-IDF, LexRank, BM25+, or LDA-extended TextRank primarily capture lexical relations [3]. Sentence embeddings instead model semantic similarity between lexically different sentences [4].

B. Keyphrase Extraction

Unsupervised keyphrase extraction includes statistical methods such as RAKE [5] and YAKE [6]; graph-based methods such as TextRank [2], TopicRank [7], MultipartiteRank [8], and WordAttractionRank [9]; and embedding-based methods. KeyBERT [10] ranks candidate n-grams by cosine similarity to the document and can apply maximal marginal relevance (MMR) [11] to reduce redundancy. Here, matching a short phrase to a document summary is an asymmetric retrieval task; a retrieval-oriented embedding model is therefore more appropriate than one optimized only for symmetric semantic textual similarity (STS).

C. Label-Space Discovery

Label-space discovery has used topic models, embedding-and-clustering methods, and LLM-assisted topic naming. Many such pipelines assume a fixed topic count or a single dominant topic per document. X-MLClass [1] uses an LLM for chunking, keyphrase generation, clustering, and iterative rare-label discovery. However, repeated raw-text processing increases token usage and latency and may produce entity-level or journalistic labels on news data. StatLM instead places a statistical encoder before the LLM so that label discovery operates on condensed keyphrases.

D. Vietnamese Processing Resources

StatLM uses pyvi [12] for word segmentation and VnCoreNLP [13] for segmentation, part-of-speech (POS) tagging, and named-entity recognition (NER). The encoder models are matched to their tasks (Table I). *dangvantuan/vietnamese-embedding* [14], a Sentence-BERT model [15] based on PhoBERT [16], is optimized for symmetric STS and reports Pearson 84.21 on *vi-stsbenchmark*. *bkai-foundation-models/vietnamese-bi-encoder* [17] is trained for asymmetric retrieval and is reported to rank first on the Vietnamese VCS benchmark.

III. THE STATLM METHOD

StatLM uses the three-layer encoder-decoder architecture in Fig. 1. Two encoder layers summarize each document and extract keyphrases; the LLM decoder then constructs the label space and assigns labels. New batches are matched to the existing label space, while unmatched documents form new clusters. Although the linguistic tools and embeddings are pretrained, no domain-specific labeled data are used for label discovery or classification.

A. Layer 1: Text Summarization

For document $D = \{s_1, \dots, s_n\}$, Layer 1 produces summary $S = \{s'_1, \dots, s'_k\}$, where $k < n$. StatLM applies TextRank with semantic edge weights rather than word overlap or other lexical similarities. Because the layer compares sentences of similar length, it uses the symmetric-STS model *dangvantuan/vietnamese-embedding* [14] with pyvi preprocessing.

The document is split into sentences and segmented with pyvi. Each sentence is encoded as $\mathbf{e}_i = \text{meanpool}(\text{PhoBERT}(s_i)) \in \mathbb{R}^{768}$, and a complete graph is built with $W_{ij} = \cos(\mathbf{e}_i, \mathbf{e}_j)$. TextRank computes

$$\text{WS}(v_i)^{(t+1)} = (1-d) + d \sum_{v_j \in \text{In}(v_i)} \frac{w_{ji}}{\sum_{v_k \in \text{Out}(v_j)} w_{jk}} \times \text{WS}(v_j)^{(t)}, \quad (1)$$

with $d = 0.85$ and convergence threshold $\varepsilon = 10^{-5}$. For $n > 5$, the $k = \max(5, \lceil 0.4n \rceil)$ highest-ranked sentences are retained and restored to their original order.

The canonical PageRank/TextRank value is fixed as $d = 0.85$ because the $(1-d)$ teleportation term only keeps the walk ergodic and the ranking is documented to be insensitive to d over $[0.8, 0.9]$. Dataset-specific fusion schemes [3] are omitted so that the retention ratio r remains the principal Layer 1 parameter.

A sentence's score accumulates its weighted semantic similarity to the rest of the document, so the roughly 60% discarded are low-centrality nodes such as redundant restatements, transitions, boilerplate, or peripheral asides. A recurring main subject is effectively preserved, whereas a secondary subject confined to a few low-centrality sentences may fall below the cut and cannot be recovered downstream. A sweep over $r \in [0.2, 0.6]$ (Table III) shows closeness to the title- and SAPO lead decreasing while topical coverage rises, so the proportional budget $k = \max(5, \lceil 0.4n \rceil)$ is adopted, consistent with embedding-based TextRank outperforming Lead- K on ROUGE (Table II).

B. Layer 2: Keyphrase Extraction

Layer 2 extracts $m = 10$ one-to-three-word keyphrases from S using KeyBERT [10] and the asymmetric retrieval model *bkai-foundation-models/vietnamese-bi-encoder* [17]. VnCoreNLP [13] supplies POS tags and NER spans. Candidate n-grams are generated within sentence boundaries, filtered by stopwords and POS tags, and supplemented with complete named-entity spans.

The procedure is: (1) preprocess with VnCoreNLP, remove stopwords, and filter by POS; (2) encode the summary into $\mathbf{E}_{doc}$; (3) generate one-to-three-word candidate n-grams within sentence fragments, adding NER entities; (4) compute

$$\cos(\mathbf{E}_{doc}, \mathbf{E}_{ngram}) = \frac{\mathbf{E}_{doc} \cdot \mathbf{E}_{ngram}}{\|\mathbf{E}_{doc}\| \|\mathbf{E}_{ngram}\|}; \quad (2)$$

Vietnamese Document Labeling Pipeline

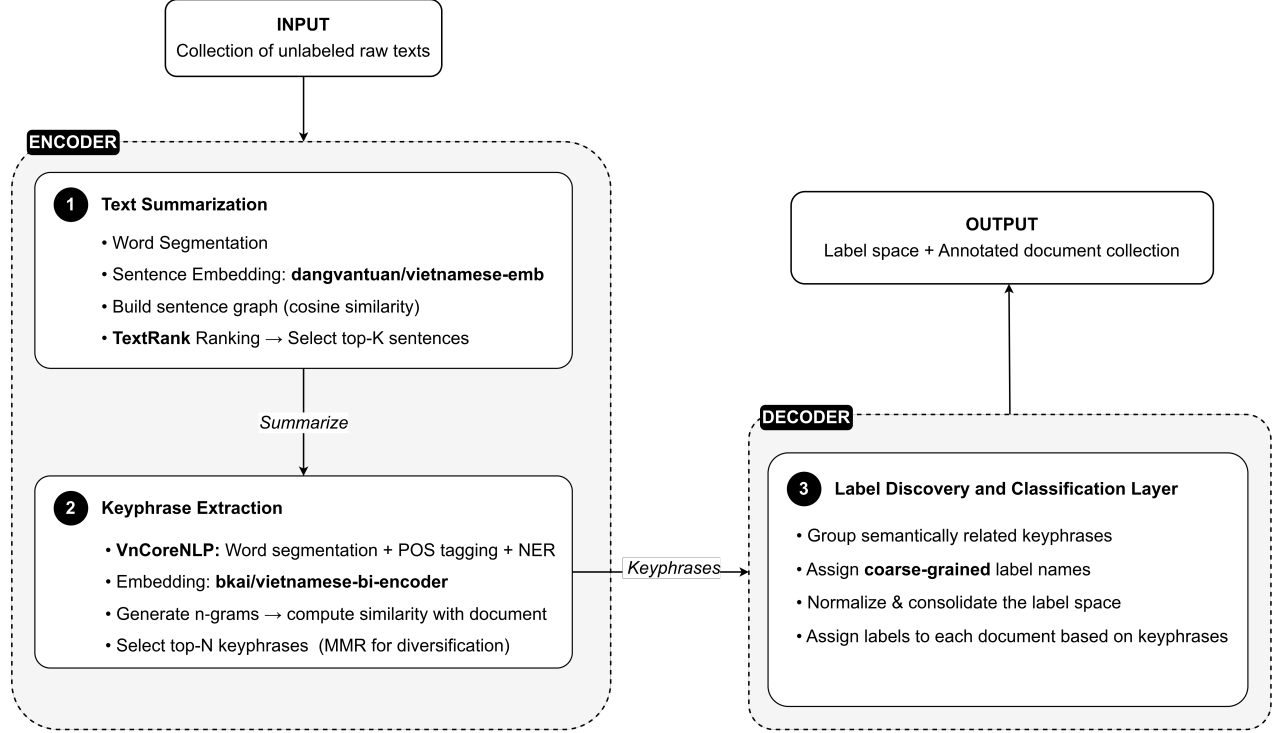


Fig. 1. StatLM architecture: two statistical encoder layers condense each document before the LLM decoder constructs, normalizes, and assigns labels.

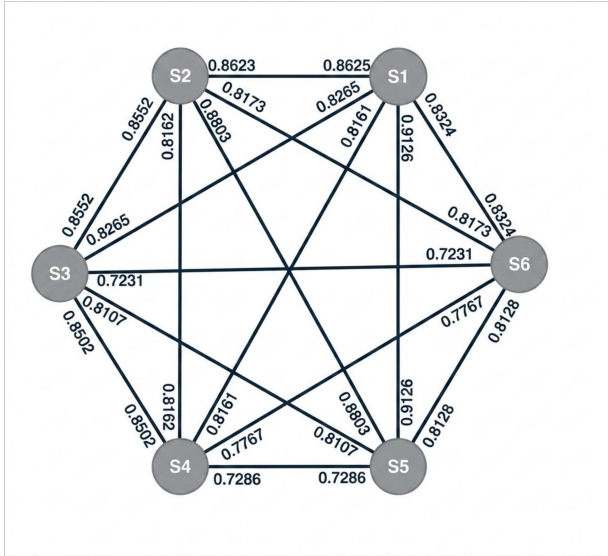


Fig. 2. Layer 1 ranks a semantic sentence graph with TextRank and retains the top- k sentences.

and (5) select keyphrases with MMR [11]:

$$\text{MMR} = \arg \max_{c \notin S} \left[\lambda \text{sim}(c, \text{doc}) - (1 - \lambda) \max_{s \in S} \text{sim}(c, s) \right], \quad (3)$$

TABLE I
EMBEDDING MODELS USED BY THE TWO ENCODER LAYERS

Criterion	Layer 1 (TextRank)	Layer 2 (KeyBERT)
Problem type	Symmetric STS	Asymmetric retrieval
Relation	Sentence to sentence	Keyphrase to document
Model	dangvantuan embedding	bkai bi-encoder
Tokenizer	pyvi	VnCoreNLP
Score	Pearson 84.21	First on VCS retrieval

with $\lambda = 0.5$ and redundancy cap $\theta = 0.7$. The neutral midpoint is used because the top-5 keyphrases must jointly span a document’s facets: emphasizing relevance yields near-duplicates, whereas emphasizing diversity admits loosely related phrases that become decoder noise. The hard cap removes a candidate whose cosine to an already selected phrase exceeds θ , preserving adjacent facets while suppressing paraphrases and head-sharing substrings. With POS- and NER-filtered candidates, selection is stable around $(\lambda, \theta) = (0.5, 0.7)$.

C. Layer 3: Classification and Labeling with the LLM

Given the keyphrase set $\mathcal{P} = \{p_1, \dots, p_N\}$ of N documents, this layer discovers the label space $\mathcal{L} = \{l_1, \dots, l_M\}$ and assigns labels. StatLM places Gemini 2.5 Flash at the decoder after filtering and condensation, so far fewer tokens reach the API and source entities or formatting are less likely to induce noisy labels. It supports new and incremental data in

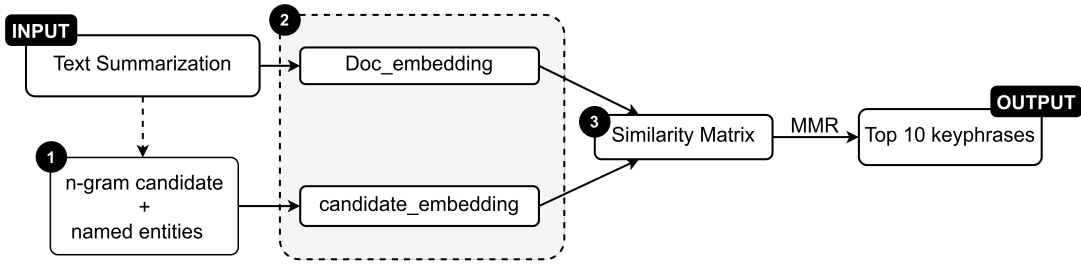


Fig. 3. Layer 2 encodes the summary and candidate n-grams independently, then uses cosine ranking and MMR to select diverse keyphrases.

three steps: (1) send only the top-5 keyphrases per document for normalization; (2) assign documents to existing clusters under a strict policy, allowing label-rename suggestions and routing non-matches to an “unassigned” list; and (3) cluster and name the unassigned documents. A System Message requests coarse-grained labels, avoids entity-like labels (e.g., “Giáo dục”/“Education” rather than “Tuyển sinh đại học 2024”/“University Admissions 2024”), minimizes label count, and merges similar topics.

IV. EXPERIMENTAL SETUP

Experiments use the *VietnamNet* dataset: 2,000 news articles evenly spread over 20 categories (100 each), each 500–800 words with a title, lead paragraph (*sapo*), and keyphrase list. All three tasks are evaluated on the full corpus. References differ by task: title + *sapo* for summarization; an LLM-generated keyphrase list (DeepSeek V4) for keyphrase extraction [18]; and the 20 original categories as a coarse-label proxy, because no absolute label-space ground truth exists [19].

All runs use a personal machine (12th-generation Intel Core i7, 16 GB RAM, no GPU); embeddings run on CPU through `sentence-transformers`, and the LLM through the Google API. Fixed parameters are: Layer 1 $d = 0.85$, $\varepsilon = 10^{-5}$, 40% retention (minimum five sentences); Layer 2 one-to-three-word n-grams, $m = 10$, $\lambda = 0.5$; and Layer 3 `gemini-2.5-flash` with temperature 0.1 for classification and 0.3 for label generation, sending at most five keyphrases per document.

Baselines. Summarization uses Lead- K , word-overlap TextRank, and TextRank+TF-IDF. Keyphrase baselines are RAKE, YAKE, TextRank-keywords, TopicRank, MultipartiteRank, WordAttractionRank, and KeyBERT with a symmetric embedding. The end-to-end *MiniLM* baseline follows X-MLClass [1] using 50-token chunks, four keyphrases per chunk, BERTopic, NLI entailment, and ten long-tail iterations; Gemini 2.5 Flash replaces the original LLM for comparability.

Metrics. Summarization uses ROUGE-1/2/L F1. Keyphrase extraction uses precision, recall, and F1 at $k = 5$ and 10 with exact or soft matching (cosine ≥ 0.7). Discovered labels are classified as Coarse, Long-tail, Entity, Meta/Noise, or Cross-topic, yielding the Coarse ratio, Noise ratio, and category

TABLE II
SUMMARIZATION ROUGE F1 SCORES ON VIETNAMNET

Method	ROUGE-1	ROUGE-2	ROUGE-L
Lead-K ($k = 5$)	0.312	0.145	0.283
Original TextRank	0.341	0.162	0.307
TextRank + TF-IDF	0.358	0.171	0.324
TextRank + embedding	0.403	0.198	0.371

coverage. End-to-end ranking is measured by

$$P@k = \frac{1}{N} \sum_{i=1}^N \frac{|C_i^{rk} \cap L_i|}{\min(k, |L_i|)}, \quad (4)$$

where C_i^{rk} is the top- k prediction and L_i is the reference set. Since $|L_i| = 1$, the measure reduces to top- k accuracy under this proxy.

V. RESULTS AND DISCUSSION

A. Summarization Evaluation (Layer 1)

Embedding-based TextRank obtains the highest ROUGE scores (Table II). Relative to word-overlap TextRank, ROUGE-1 F1 increases by approximately 18% (0.341 to 0.403), and ROUGE-2 F1 increases by approximately 22% (0.162 to 0.198). The results support semantic rather than lexical edge weights for the sentence graph.

Sweeping $r \in [0.2, 0.6]$ (Table III), semantic closeness to the title+*sapo* lead (SemSim) falls while topical coverage (KP-Cov) rises. Both coverage and token cost grow with r , so $r = 0.4$ is a balanced operating point: it retains sufficient coverage for open-world discovery without inflating the LLM token load.

B. Keyphrase-Extraction Evaluation (Layer 2)

StatLM’s KeyBERT leads on every metric (Table IV). The gap between the two KeyBERT variants (F1@10: 30.10 vs. 25.51, about 18%) confirms the role of *asymmetric embedding* in bridging short keyphrases and long documents. WordAttractionRank leads the graph group (F1@10 = 25.21) but trails KeyBERT because it lacks a redundancy mechanism such as MMR.

TABLE III
LAYER-1 RETENTION SWEEP; $r=0.4$ BALANCES LEAD SIMILARITY AND TOPICAL COVERAGE

Retention ratio r	Compression ratio	Semantic similarity $\downarrow$	Keyphrase coverage $\uparrow$
0.20	0.39	0.510	0.266
0.25	0.41	0.506	0.275
0.30	0.45	0.501	0.287
0.35	0.49	0.496	0.298
0.40	0.53	0.490	0.306
0.45	0.57	0.483	0.316
0.50	0.61	0.477	0.324
0.55	0.66	0.470	0.334
0.60	0.70	0.464	0.340

TABLE IV
KEYPHRASE-EXTRACTION RESULTS ON VIETNAMNET (IN %)

Method	P@5	R@5	F1@5	P@10	R@10	F1@10
RAKE	17.32	10.84	13.33	11.50	13.70	12.50
YAKE	20.15	12.67	15.56	14.45	17.50	15.83
TextRank (keywords)	19.23	12.31	15.01	13.21	16.53	14.68
TopicRank	25.02	16.45	19.85	19.34	24.52	21.62
MultipartiteRank	26.83	17.67	21.31	20.47	26.26	23.01
WordAttractionRank	28.49	17.43	21.63	23.16	27.66	25.21
KeyBERT + sym. emb.	28.56	17.41	21.63	23.48	27.93	25.51
KeyBERT (StatLM)	32.21	21.64	25.89	26.64	34.58	30.10

C. Label-Space and Classification Evaluation (Layer 3)

Five evaluators independently sorted 72 labels (26 StatLM, 46 MiniLM) into the five groups under a standardized guideline. Fleiss' $\kappa = 0.806$ ($\bar{P} \approx 0.866$, $\bar{P}_e \approx 0.307$), an *almost perfect* level according to Landis and Koch [20], confirms reliable criteria.

MiniLM produces 18 Entity or Meta/Noise labels, corresponding to a 39.1% Noise ratio, and covers 15 of 20 editorial categories with Coarse labels. StatLM produces one Meta/Noise label, no Entity or Long-tail labels, and covers 19 categories, yielding a 3.8% Noise ratio and a 76.9% Coarse ratio (Table V). Its five Cross-topic labels reduce fragmentation but may also obscure distinctions needed downstream. Thus, 19/20 coverage denotes a compatible coarse label, not exact recovery of the editorial taxonomy.

StatLM obtains $P@1 = 0.71$ and $P@3 = 0.84$, compared with 0.49 and 0.63 for MiniLM. Under the single-label proxy, these measures are top-1 and top-3 accuracy. They do not capture additional valid document facets that are absent from the reference. Consequently, the Coarse ratio, Noise ratio, category coverage, and ranking scores should be interpreted jointly. Optimizing only the Coarse ratio could favor an over-merged taxonomy, whereas optimizing only coverage could produce redundant or entity-specific labels. A deployment-oriented evaluation should additionally assess label stability across batches, hierarchical consistency, and the frequency with which Cross-topic labels require manual refinement.

TABLE V
MINILM AND STATLM LABEL-SPACE QUALITY

Label group	MiniLM		StatLM	
	No.	Ratio	No.	Ratio
Coarse	15	32.6%	20	76.9%
Long-tail	8	17.4%	0	0.0%
Entity	6	13.0%	0	0.0%
Meta/Noise	12	26.1%	1	3.8%
Cross-topic	5	10.9%	5	19.2%
Total	46	100%	26	100%
Noise ratio	39.1%		3.8%	
Categories covered (Coarse)	15/20 (75%)		19/20 (95%)	

TABLE VI
EFFECT OF EACH LAYER ON THE FINAL LABEL-SPACE QUALITY (L1: ROUGE-1 F1; L2: F1@10; L3: COARSE / NOISE / CATEGORIES COVERED).

Variant	L1	L2	L3 (Coarse/Noise/Cov.)
Full StatLM	0.403	30.10	76.9% / 3.8% / 19/20
L1 $\rightarrow$ orig. TextRank	0.341	28.5	68% / 10% / 17/20
Drop L1	–	27.0	65% / 15% / 16/20
L2 $\rightarrow$ symmetric	0.403	25.51	70% / 8% / 17/20
Drop L2	0.403	–	55% / 20% / 14/20

D. Performance Evaluation

On the evaluation platform, StatLM reduces mean processing time from approximately 40 to 10 seconds per document, LLM input from approximately 900 to 30 tokens per document, and API calls from approximately eight to one. These figures correspond to a 75% time reduction and about 30 times fewer tokens. Absolute latency may differ under other hardware and API conditions.

E. Ablation and Discussion

Table VI reports the ablation results for the main components of StatLM. Each configuration was evaluated on the complete VietnamNet corpus of 2,000 documents over three independent runs, while the remaining components, prompts, model settings, and evaluation procedure were kept unchanged. The reported values are arithmetic means over the three runs, rounded to the displayed precision.

Replacing the embedding-based TextRank in Layer 1 with the original lexical variant reduces ROUGE-1 F1 from 0.403 to 0.341 and degrades the subsequent keyphrase and label-space metrics. Removing Layer 1 further increases the noise-label ratio, indicating that summarization serves not only to reduce document length but also to suppress redundant and peripheral content before keyphrase extraction. The largest degradation occurs when Layer 2 is removed: the coarse-label ratio decreases from 76.9% to 55.0%, the noise-label ratio rises from 3.8% to 20.0%, and category coverage decreases from 19/20 to 14/20. This result shows that sentence-level

summaries alone are insufficient for producing a compact, coarse-grained label space.

Replacing the asymmetric bi-encoder in Layer 2 with a symmetric sentence-embedding model also reduces F1@10 from 30.10 to 25.51 and degrades the final label space. Overall, the results indicate that the two encoder layers make distinct but complementary contributions: Layer 1 improves the semantic concentration of the input, whereas Layer 2 performs topic-oriented condensation and redundancy control before LLM-based generalization. The full configuration therefore provides empirical support for the proposed “clean first, generalize later” design.

VI. CONCLUSION

This paper presented StatLM, a hybrid pipeline for open-world zero-shot multi-label classification of Vietnamese text. Embedding-based TextRank and asymmetric KeyBERT condense each document before an LLM constructs and assigns labels. On the VietnamNet corpus, StatLM achieved a 76.9% Coarse ratio, a 3.8% Noise ratio, and coverage of 19 of 20 categories. Relative to the X-MLClass-based MiniLM baseline, it reduced mean processing time by approximately 4 times, LLM token usage by approximately 30 times, and API calls from approximately eight to one.

The results indicate complementary roles for the statistical encoder and generative decoder: the encoder controls retention and redundancy, while the decoder maps the compact representations to a normalized label space. However, omitted information cannot be recovered downstream, the complete sentence graph requires $O(n^2)$ pairwise similarities, and LLM outputs may vary across runs. The evaluation also uses an LLM-generated keyphrase reference and one editorial category per article for several component variants. The reported latency and token counts were obtained under one hardware and API configuration and do not constitute a platform-independent cost model. These factors, together with the news-only corpus and Vietnamese-specific resources, limit internal and external validity. The study therefore does not claim universal superiority over full-LLM pipelines. Its balanced news corpus and language-specific resources limit transfer, while the LLM-generated reference and the limited number of ablation runs may introduce bias and warrant confirmation through additional complete runs.

Future work will incorporate structural cues into the sentence graph and evaluate the method in additional domains. We will also develop human-annotated Vietnamese benchmarks for keyphrase extraction and multi-label classification. Complete component ablations and repeated LLM runs will be used to report variance and statistical uncertainty. A confidence-based fallback will reintroduce the full document when encoder coverage or decoder confidence is insufficient, reducing the open-loop behavior of the current cascade.

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